

COMS 4705 Paper Proposal

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Introduction

Project Proposal: Evaluating Chain-of-Thought Prompting and Self-Consistency in LLMs

For my paper, I would like to propose the idea of exploring how test-time computation techniques would help to enhance the multi-step reasoning capabilities of LLMs. Specifically, I would like to implement and evaluate the foundational Chain-of-Thought prompting methods alongside with its series, and compare it against the standard method of prompting Wei et al. [2022]. To extend the project a little further, I may also test Self-Consistency method, which replace greedy decoding by sampling multiple reasoning paths and using majority voting to reach a final answer Wang et al. [2023].

Implementation

In order to implement this idea, I would use the Hugging Face transformers library with a small open-weight model (such as a smaller variant of Llama 3) to make sure that the computation is easily tractable. I would use four approaches (Standard IO, Pure CoT, and CoT + Self-Consistency) on a given benchmark (to be decided). The coding portion of this project involves writing the reference scripts for few-shot prompting templates, generating responses from the model, and the logic for majority voting that is required by Self-Consistency. **GSM8K** may be the benchmark that I'll use as it is a gold standard to measure CoT and Self-Consistency.

Final Paper of the Project

The final paper will both explain the theoretical frameworks behind CoT and Self-Consistency and how they could help in reducing the cognitive load at each reasoning step. I will further compare my implementation results with the theory and write up an analysis alongside with the comparison.

References

- Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V. Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-Consistency Improves Chain of Thought Reasoning in Language Models. In *International Conference on Learning Representations*, 2023. URL <https://arxiv.org/abs/2203.11171>.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. In *Advances in Neural Information Processing Systems 35*, 2022. URL <https://arxiv.org/abs/2201.11903>.